ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme	:	BACHELOR OF COMPUTER ENGINEERING TECHNOLOGY (COMPUTER SYSTEMS) WITH HONOURS
Course Code & Name	:	IPB 49906 FINAL YEAR PROJECT 2
Course Lecturer(s)	:	Ts. DIYANA BT AB KADIR (FYP2 BCE COORDINATOR)
Submission Title	:	PROGRESS REPORT
Deadline	:	Day 10 Month MAY Year 2026 Time 11:50PM
Penalties	:	• 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted. • Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (**√**) each box below if you agree:

√	I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
√	This submission is my own, unless indicated with proper referencing.
√	This submission has not been previously submitted or published.
√	This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
15/4/2026	MD EJAJ MAHMUD	52222222123

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)
15/4/2026	IPB 49906	Progress Report 3	52222222123

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RUBRIC FOR PROJECT PROGRESS

COURSE CODE	IPB 49906	TOTAL MARKS (50 MARKS)
COURSE NAME	FINAL YEAR PROJECT 2	
COURSE SEMESTER/ YEAR	MARCH 2026	
STUDENT NAME	MD EJAJ MAHMUD	
STUDENT ID	52222222123	
ASSESSMENT DATE	WEEK 1 – WEEK 11	Smart Recycling: Reward-Based Reverse Vending Machine Using Proximity Sensor Detection for Plastic Waste Management
ASSESSMENT DUE	29 MAY 2026 4PM	
NAME OF SUPERVISOR	Dr. Grad. Engr. Hannah Binti Sofian	

NO.	CRITERIA	Description	WEIGHTAGE	SCORE 1-10	ACTUAL SCORE
1	**Consultation with Supervisor**	Meet Supervisor with significant progress	1		
2	**Timelines**	Record of research progress with respect to the Timelines	1		
3	**Research progress**	Show progress in research with related information and resources	1		
4	**Proposed Idea**	Ability to apply higher order thinking	1		
5	**Self-driven & Independency**	Ability to perform independently	1		
				TOTAL SCORE (50 MARKS)	

PROGRESS REPORT NO. 2

Student's Name: MD EJAJ MAHMUD

FYP Code : IPB49906

Semester	07		Week	9
Meeting No.	01		Date	20/05/2026

Progress (fill in by Student)
The main components arrived in Week 6, so I was able to move from simulation into actual hardware work. I started by testing each part individually before connecting everything together. I connected the Arduino Mega 2560 to the breadboard and ran a basic blink test to confirm it was working correctly. After that, I wired the 16x2 LCD display using analog pins A0 to A5 with the potentiometer controlling the contrast. The display showed the correct startup text on the first attempt. I then connected and tested the inductive proximity sensor on pin 27 and the capacitive proximity sensor on pin 31. Using the Serial Monitor, I confirmed that the inductive sensor correctly reads LOW when no metal is nearby and HIGH when metal is detected. The capacitive sensor returned HIGH consistently when a plastic bottle was placed near it. The IR sensor on pin 49 was connected and tested by passing a small object through the beam path. It successfully triggered the expected signal change, confirming that the bottle gate detection will work as designed. The ultrasonic sensor HC-SR04 was connected to trigger pin 51 and echo pin 53. Distance readings were verified using Serial Monitor at several positions. Readings at the 30 cm threshold were consistent with the defined bin-full condition in the code. I uploaded the full Arduino sketch (Bottle_WasteCode.ino) to the board. The system responded with correct LCD messages, buzzer output, and LED feedback under the tested input conditions. Chapter 3 (Methodology) of the FYP2 report has been updated to include the actual pin assignment table, hardware connection description, and annotated circuit schematic from Proteus. Appendix 1 shows the initial hardware wiring setup for Progress Report 3. **Appendix 1: Project Simulation** Current Proteus simulation prepared while waiting for the remaining hardware items.

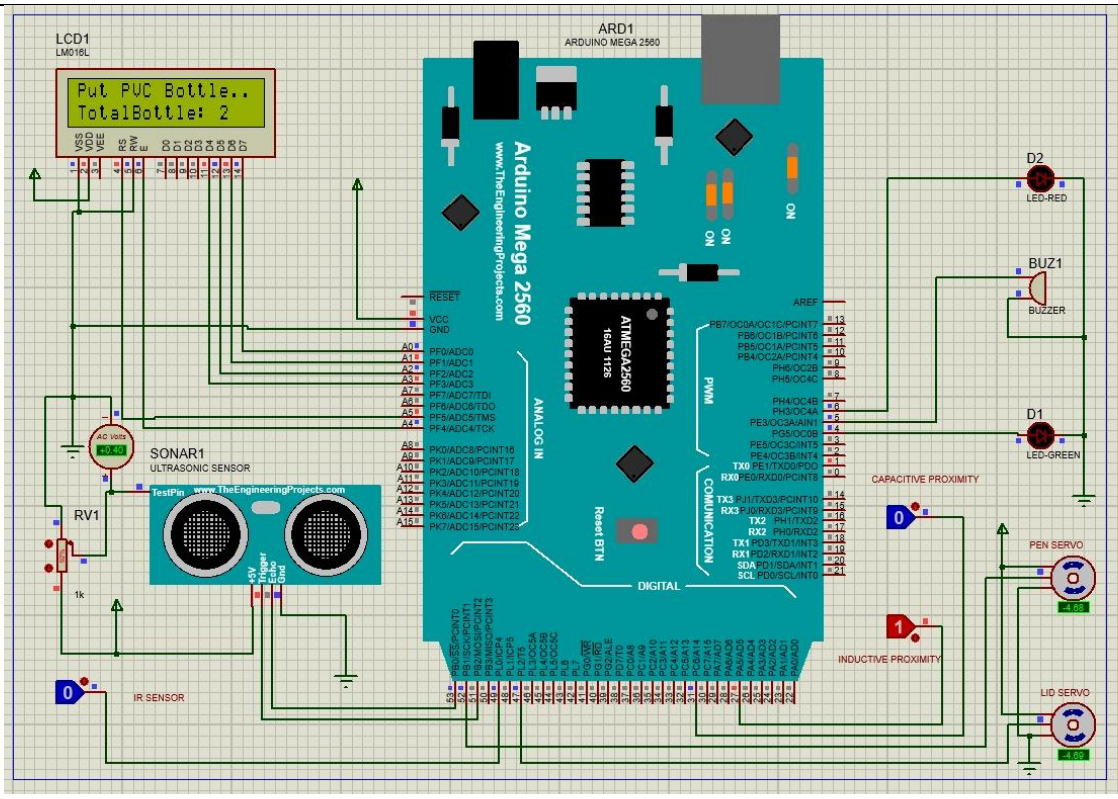

Figure A1. Initial hardware wiring setup prepared for Progress Report 3.

Figure A2: Equipment Arrived

Comments (by Supervisor)	Signature and Name
